

Problem

An audio amplifier with a resistance of $9\ \Omega$ supplies power to a speaker. What should be the resistance of the speaker for maximum power to be delivered?

Step-by-step solution

Step 1 of 2

According to the Maximum power transfer theorem, any circuit supplies maximum power to the load when the connected load is equal to the complex conjugate of the equivalent impedance of the circuit.

$$P_L = P_{L(\max)}$$

when,

$$Z_L = Z_S^*$$

$$R_L + jX_L = R_S - jX_S$$

Step 2 of 2

The audio amplifier acts as a source and speaker acts as a load.

$$\begin{aligned} R_S &= R_{\text{audio amp}} \\ &= 9\ \Omega \end{aligned}$$

For maximum power flow,

$$\begin{aligned} R_{\text{speaker}} &= R_{\text{audio amp}} \\ &= 9\ \Omega \end{aligned}$$

Therefore, the resistance of the speaker is $R_{\text{speaker}} = 9\ \Omega$.